

## MOYALE, Kenya

WMO: 636190

Lat: 3.533N

Lon: 39.050E

Elev: 3599

StdP: 12.88

Time Zone: 3.00 (E03)

Period: 90-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 61.1       | 62.3 | 36.5                          | 36.2 | 81.8 | 40.1 | 41.6 | 80.4 | 24.6                  | 68.9 | 21.9 | 69.3 | 7.5                   | 180  | 0.285 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 20.0                   | 90.8            | 63.8 | 89.4 | 63.9 | 87.8 | 64.0 | 71.6                | 78.7 | 70.8 | 77.9 | 70.1 | 77.1 | 9.9                  | 90   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 69.7                            | 125.0 | 74.6 | 68.6 | 120.2 | 73.3 | 68.2 | 118.4 | 72.8 | 37.9          | 78.9 | 37.1 | 77.6 | 36.5 | 77.8 | 80.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 21.4          | 18.2 | 15.5 | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 73.3   | 77.2  | 79.1  | 77.3  | 73.9  | 71.9  | 70.3  | 68.8  | 69.6  | 72.0  | 72.6  | 72.6  | 74.2 |
|                                                                               | DBStd       | 3.85   | 2.19  | 2.55  | 2.84  | 2.32  | 1.88  | 2.16  | 2.43  | 2.27  | 2.34  | 2.62  | 1.91  | 2.25 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 3      | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 1     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 8490   | 843   | 816   | 847   | 717   | 680   | 610   | 582   | 607   | 660   | 701   | 677   | 750  |
|                                                                               | CDD65       | 3018   | 378   | 396   | 382   | 267   | 215   | 160   | 119   | 143   | 210   | 236   | 227   | 285  |
|                                                                               | CDH74       | 17321  | 3409  | 3947  | 3078  | 1070  | 527   | 320   | 224   | 390   | 862   | 853   | 832   | 1809 |
|                                                                               | CDH80       | 4878   | 1289  | 1642  | 1059  | 145   | 20    | 15    | 7     | 15    | 98    | 115   | 78    | 395  |
| Wind                                                                          | WSAvg       | 6.6    | 7.2   | 7.0   | 6.2   | 5.2   | 5.5   | 6.7   | 8.4   | 9.0   | 7.9   | 5.9   | 5.2   | 5.5  |
| Precipitation                                                                 | PrecAvg     | 27.6   | 0.7   | 0.8   | 2.4   | 7.7   | 5.3   | 0.7   | 0.7   | 0.6   | 0.8   | 4.1   | 3.8   | 1.6  |
|                                                                               | PrecMax     | 49.6   | 6.3   | 3.9   | 7.2   | 25.6  | 18.5  | 2.2   | 2.4   | 2.4   | 3.3   | 13.2  | 10.9  | 6.3  |
|                                                                               | PrecMin     | 14.0   | 0.0   | 0.0   | 0.0   | 2.0   | 0.8   | 0.0   | 0.0   | 0.0   | 0.0   | 0.4   | 0.7   | 0.0  |
|                                                                               | PrecStd     | 8.0    | 1.3   | 0.9   | 2.0   | 5.4   | 3.2   | 0.6   | 0.5   | 0.5   | 0.9   | 3.1   | 2.6   | 1.5  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 91.6  | 93.2  | 91.7  | 85.3  | 82.2  | 81.3  | 80.7  | 82.1  | 84.5  | 85.6  | 84.4  | 88.0 |
|                                                                               |             | MCWB   | 64.5  | 63.9  | 63.1  | 68.6  | 69.4  | 66.0  | 64.3  | 64.4  | 65.4  | 67.4  | 66.0  | 63.8 |
|                                                                               | 2%          | DB     | 89.7  | 91.3  | 89.6  | 83.6  | 80.4  | 79.0  | 78.6  | 80.2  | 82.5  | 82.5  | 82.4  | 86.0 |
|                                                                               |             | MCWB   | 63.8  | 63.7  | 65.0  | 68.5  | 68.9  | 65.3  | 63.9  | 63.7  | 65.0  | 66.7  | 66.7  | 63.9 |
|                                                                               | 5%          | DB     | 88.1  | 89.8  | 87.8  | 81.9  | 78.7  | 77.2  | 76.7  | 78.3  | 80.7  | 80.6  | 80.7  | 84.2 |
|                                                                               |             | MCWB   | 63.5  | 63.4  | 65.1  | 68.6  | 68.6  | 64.7  | 63.6  | 63.2  | 64.4  | 66.3  | 67.1  | 64.3 |
|                                                                               | 10%         | DB     | 86.5  | 88.5  | 86.1  | 80.2  | 77.1  | 75.4  | 74.7  | 76.3  | 78.9  | 78.7  | 79.0  | 82.3 |
|                                                                               |             | MCWB   | 63.3  | 63.6  | 65.3  | 68.9  | 68.3  | 64.5  | 63.2  | 62.9  | 63.9  | 66.5  | 67.2  | 65.2 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 70.8  | 70.4  | 71.9  | 72.5  | 72.4  | 68.5  | 70.1  | 69.4  | 68.3  | 70.9  | 71.8  | 71.6 |
|                                                                               |             | MCDB   | 82.6  | 83.2  | 80.0  | 79.1  | 78.6  | 76.4  | 73.7  | 72.9  | 79.3  | 77.9  | 78.2  | 80.0 |
|                                                                               | 2%          | WB     | 69.0  | 68.9  | 70.6  | 71.2  | 71.0  | 67.1  | 66.7  | 65.7  | 66.9  | 69.6  | 70.5  | 70.2 |
|                                                                               |             | MCDB   | 80.4  | 81.7  | 78.9  | 78.2  | 76.6  | 74.7  | 72.3  | 75.2  | 77.6  | 76.2  | 76.6  | 78.3 |
|                                                                               | 5%          | WB     | 67.8  | 67.8  | 69.7  | 70.6  | 70.0  | 66.2  | 65.2  | 64.5  | 66.0  | 68.8  | 69.4  | 69.1 |
|                                                                               |             | MCDB   | 78.9  | 81.0  | 78.4  | 77.3  | 75.6  | 73.7  | 72.1  | 74.2  | 76.5  | 75.3  | 75.4  | 76.8 |
|                                                                               | 10%         | WB     | 66.6  | 66.9  | 68.9  | 69.8  | 69.2  | 65.5  | 64.3  | 63.7  | 65.0  | 68.1  | 68.7  | 68.2 |
|                                                                               |             | MCDB   | 78.8  | 80.3  | 77.8  | 76.3  | 74.5  | 72.7  | 71.6  | 73.3  | 75.5  | 74.4  | 74.8  | 75.6 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 20.3  | 20.0  | 16.9  | 12.4  | 11.2  | 12.0  | 11.4  | 12.9  | 14.5  | 12.9  | 13.2  | 17.4 |
|                                                                               |             | MCDBR  | 22.4  | 21.3  | 19.5  | 15.0  | 13.2  | 14.4  | 14.7  | 16.0  | 17.1  | 16.1  | 16.2  | 19.8 |
|                                                                               |             | MCWBR  | 8.1   | 7.3   | 6.8   | 5.5   | 5.2   | 5.2   | 5.7   | 5.6   | 5.7   | 5.6   | 6.3   | 7.0  |
|                                                                               | 5% WB       | MCDBR  | 18.4  | 19.0  | 15.4  | 12.6  | 11.3  | 12.3  | 11.9  | 13.4  | 15.1  | 12.9  | 12.4  | 15.7 |
|                                                                               |             | MCWBR  | 7.4   | 7.5   | 6.0   | 5.8   | 5.4   | 5.1   | 6.4   | 5.8   | 6.0   | 5.7   | 6.0   | 7.5  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.338  | 0.357 | 0.379 | 0.380 | 0.381 | 0.447 | 0.454 | 0.406 | 0.381 | 0.373 | 0.367 | 0.347 |      |
|                                                                               | taud        | 2.558  | 2.490 | 2.455 | 2.504 | 2.504 | 2.238 | 2.204 | 2.369 | 2.449 | 2.515 | 2.519 | 2.552 |      |
|                                                                               | Ebn at Noon | 313    | 308   | 299   | 293   | 286   | 263   | 263   | 282   | 294   | 299   | 301   | 308   |      |
|                                                                               | Edh at Noon | 33     | 36    | 37    | 35    | 34    | 43    | 45    | 39    | 37    | 35    | 34    | 33    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 2076   | 2144  | 2033  | 1707  | 1607  | 1431  | 1405  | 1588  | 1792  | 1595  | 1678  | 1898  |      |
|                                                                               | RadStd      | 106    | 97    | 132   | 81    | 64    | 118   | 106   | 84    | 77    | 67    | 69    | 133   |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page